

HDOC DAY-8 REPORT

C Programming Activity

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Question 1

Write a C program to find the volume and surface area of a wooden block shaped like a cuboid with a cylindrical hole through it.

Step 1 - Problem Understanding

Assumption: the cylindrical hole passes completely through the height of the cuboid, and its circular openings lie on the top and bottom faces.

- Inputs: cuboid length L (cm), breadth B (cm), height H (cm), and cylindrical hole radius r (cm).
- Outputs: remaining volume (cubic cm) and total exposed surface area (square cm), including the inner curved surface of the hole.
- Volume formula: $V = L \times B \times H - \pi \times r^2 \times H$.
- Surface area formula: $SA = 2(LB + LH + BH) - 2\pi r^2 + 2\pi rH$.
- Validity condition: the hole must fit inside the top/bottom face, so $2r \leq \min(L, B)$.

Step 2 - Test Case Table

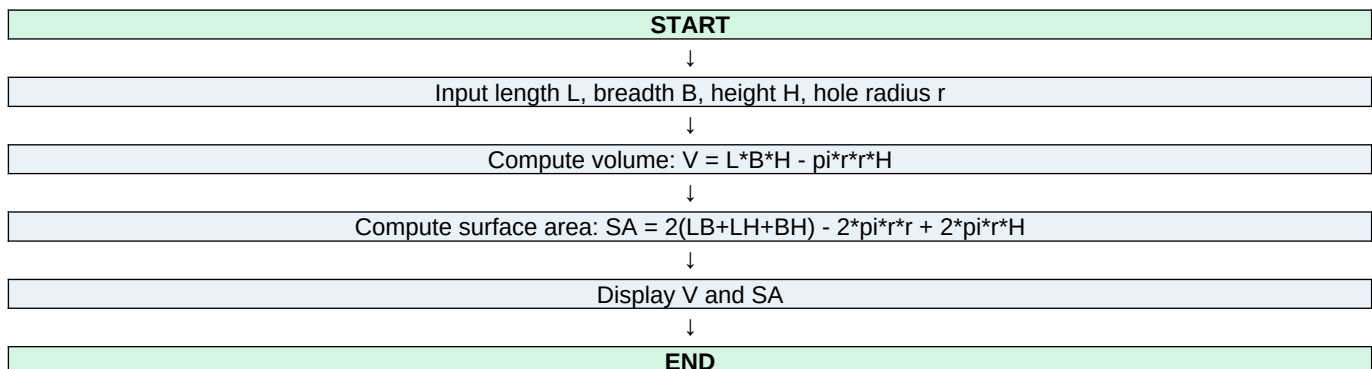
Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	$L=10, B=8, H=6, r=2$	$V=404.60 \text{ cm}^3$ $SA=426.27 \text{ cm}^2$	$V=404.60 \text{ cm}^3$ $SA=426.27 \text{ cm}^2$	Pass
2	$L=12, B=10, H=5, r=1.5$	$V=564.66 \text{ cm}^3$ $SA=492.99 \text{ cm}^2$	$V=564.66 \text{ cm}^3$ $SA=492.99 \text{ cm}^2$	Pass
3	$L=20, B=15, H=10, r=3$	$V=2717.26 \text{ cm}^3$ $SA=1431.95 \text{ cm}^2$	$V=2717.26 \text{ cm}^3$ $SA=1431.95 \text{ cm}^2$	Pass

Step 3 - Algorithm and Evaluation

- Start.
- Read length L , breadth B , height H and hole radius r .
- Calculate cuboid volume: $L \times B \times H$.
- Calculate cylindrical-hole volume: $\pi \times r \times r \times H$.
- Calculate remaining volume: cuboid volume - hole volume.
- Calculate original cuboid surface area: $2(LB + LH + BH)$.
- Subtract the two circular areas removed from the top and bottom: $2\pi r^2$.
- Add the inner curved surface area of the hole: $2\pi rH$.
- Display volume and surface area.
- Stop.

Evaluation: For all three prepared test cases, applying the algorithm gives the same values as the expected outputs; therefore the algorithm passes all test cases.

Step 4 - Flowchart and Evaluation



Flowchart evaluation: Each test case follows the same sequence from input to calculation and output, and produces the expected results. Hence the flowchart logic passes all three test cases.

Step 5 - C Program

```
#include <stdio.h>
#define PI 3.141592653589793

int main() {
    double length, breadth, height, radius;
    double volume, surfaceArea;

    printf("Enter length, breadth, height and hole radius (cm): ");
    scanf("%lf %lf %lf %lf", &length, &breadth, &height, &radius);

    volume = length * breadth * height - PI * radius * radius * height;
    surfaceArea = 2 * (length * breadth + length * height + breadth * height)
        - 2 * PI * radius * radius + 2 * PI * radius * height;

    printf("Volume = %.2f cubic cm\n", volume);
    printf("Surface Area = %.2f square cm\n", surfaceArea);
    return 0;
}
```

Step 6 - Compilation, Execution and Code Evaluation

The program was compiled using GCC. Compilation completed successfully with no errors. The program was then executed for all three test cases; actual outputs matched the expected outputs, so all tests passed.

Test Case	Compilation	Execution Result
1	Successful	Pass
2	Successful	Pass
3	Successful	Pass

Step 7 - Compilation and Execution Evidence

```
$ gcc day8_q1.c -lm -o day8_q1
Compilation successful (no errors).

$ ./day8_q1
Enter length, breadth, height and hole radius (cm): 10 8 6 2
Enter length, breadth, height and hole radius (cm): Volume = 404.60 cubic cm
Surface Area = 426.27 square cm
```

Note: The terminal-style block above records the compilation and one execution run. You may replace or supplement it with your own IDE/terminal screenshots before GitHub submission if your instructor specifically requires image screenshots.

Question 2

Write a C program to find the cost of bordering a semicircular window when radius and border rate per centimeter are given.

Step 1 - Problem Understanding

Assumption: bordering means the complete boundary of the semicircular window, including the curved semicircular arc and the straight diameter.

- Inputs: radius r (cm) and border rate R (cost per cm).
- Outputs: total border length (cm) and total bordering cost.
- Semicircular arc length = πr .
- Diameter = $2r$.
- Total border length $P = \pi r + 2r$.
- Cost $C = P \times R$.

Step 2 - Test Case Table

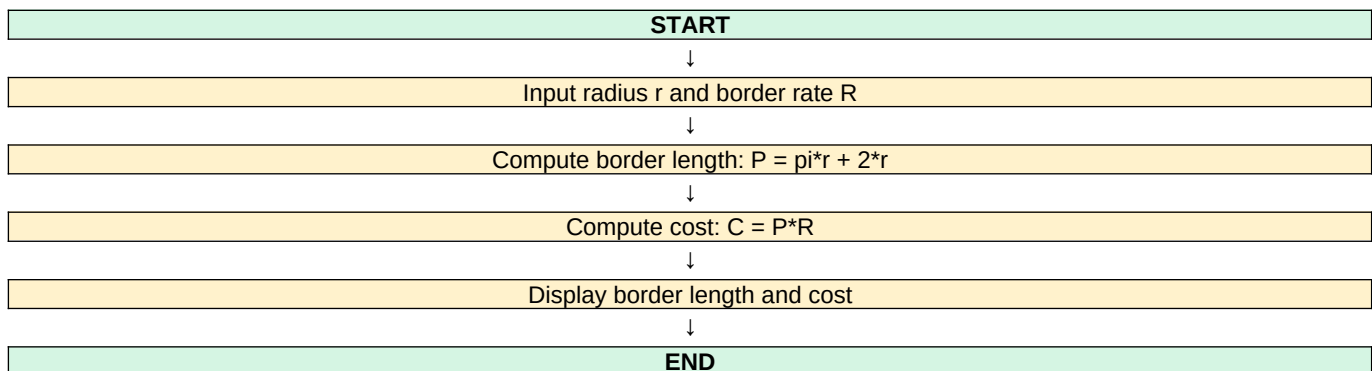
Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	$r=10$, $rate=5$	Length=51.42 cm Cost=257.08	Length=51.42 cm Cost=257.08	Pass
2	$r=7$, $rate=12.5$	Length=35.99 cm Cost=449.89	Length=35.99 cm Cost=449.89	Pass
3	$r=15$, $rate=8$	Length=77.12 cm Cost=616.99	Length=77.12 cm Cost=616.99	Pass

Step 3 - Algorithm and Evaluation

- Start.
- Read radius r and border rate R per centimeter.
- Calculate semicircular arc length: $\pi \times r$.
- Calculate diameter: $2 \times r$.
- Calculate total border length: $\pi \times r + 2 \times r$.
- Calculate cost: border length $\times$ rate.
- Display border length and cost.
- Stop.

Evaluation: The calculated border length and cost match the expected values for all three test cases. Therefore the algorithm passes all test cases.

Step 4 - Flowchart and Evaluation



Flowchart evaluation: The input, calculation and output sequence gives the expected result for each prepared test case. Hence the flowchart passes all three test cases.

Step 5 - C Program

```
#include <stdio.h>
#define PI 3.141592653589793

int main() {
    double radius, rate, borderLength, cost;

    printf("Enter radius (cm) and border rate per cm: ");
    scanf("%lf %lf", &radius, &rate);

    borderLength = PI * radius + 2 * radius;
    cost = borderLength * rate;

    printf("Border Length = %.2f cm\n", borderLength);
    printf("Bordering Cost = %.2f\n", cost);
    return 0;
}
```

Step 6 - Compilation, Execution and Code Evaluation

The program was compiled using GCC with no errors and executed for all three test cases. Each actual output matched its expected output, so all tests passed.

Test Case	Compilation	Execution Result
1	Successful	Pass
2	Successful	Pass
3	Successful	Pass

Step 7 - Compilation and Execution Evidence

```
$ gcc day8_q2.c -o day8_q2
Compilation successful (no errors).

$ ./day8_q2
Enter radius (cm) and border rate per cm: 10 5
Enter radius (cm) and border rate per cm: Border Length = 51.42 cm
Bordering Cost = 257.08
```

Note: The terminal-style block above records the compilation and one execution run. You may replace or supplement it with your own IDE/terminal screenshots before GitHub submission if your instructor specifically requires image screenshots.

Conclusion

Both C programs were designed, tested and evaluated using three test cases each. The algorithms and flowcharts produced the expected results, and both programs compiled successfully and passed all prepared test cases.